

People keep the moment and lose the date, but Photos indexes photos, not moments

Strategic Goal (Brief p. 2)

"Increase the percentage of users who successfully retrieve a photo they remember but cannot precisely describe."

Core Finding Across Two Independent Research Methods

77.1% of observed search failures occur before recovery can help: the query clue is misread (35.4%) or the photo never surfaces in results (41.7%). (§B)

Survey (n=15) and user testing (n=6) agree: people keep *roughly when* and the event, but lose the calendar date. 4 of 5 real-app users strip time into bare nouns, and 100% of real searches failed. (§G1, §G2)

85,140

Public posts analyzed across 4 platforms in discovery engine

+0.790

Recall@20 lift at L3 (0.172 → 0.962) via episode re-entry

77.1%

Failures at interpretation & surfacing before recovery

4.0 / 5

Ease rating vs Google Photos across 6 target segment testers

What Was Built & Deployed Live

1. Discovery Engine: Structured pipeline turning 85,140 posts into 144 specific attempts with blind two-model audit.

2. Memory Trails MVP: Episode-first re-entry prototype inside Google Photos. Resolves vague dates, clusters moments, and explains matches with soft scoring.



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sister's graduation ✓ cake ✓

[retrieval-discovery-engine.vercel.app ↗](https://retrieval-discovery-engine.vercel.app)

[memory-trails-v2.vercel.app ↗](https://memory-trails-v2.vercel.app)

Retrieval breaks most at interpretation and surfacing, and almost never at recovery

Level 0 North Star Metric: URR (User Retrieval Rate)

Share of users with ≥1 vague-memory retrieval task in 28 days who successfully reach the photo. A/B splits by user.

$$URR = Expression \times [1 - (1 - Interpretation \times Surfacing \times Recognition)^{\bar{n}}]$$

Product Outcome	The Brief's Guiding Question	Engine (144)	Survey (15)	Observed User Behaviour
Expression	Is the user unable to express what they remember?	2 (1.4%)	3 (20%)	Does not know what to type; 2 of 3 never typed a search
Interpretation ★	Does Google Photos fail to understand clues?	51 (35.4%)	3 (20%)	Adds year, rewords: "You must now specify the year"
Surfacing ★	Is the photo in results at all? (Our term)	60 (41.7%)	5 (33%)	Scrolls grid of reasonable photos; target photo absent
Recognition	Are potentially relevant results hard to evaluate?	2 (1.4%)	2 (13%)	Can't confirm; 7 ended unsure; wants trip/context photos
Recovery	Does the user struggle to refine a failed search?	1 (0.7%)	1 (7%)	Retries: 2–3 searches for 9 of 13 → $\bar{n}$ in formula

Parallel User Path: Browse (Outside Formula)

Survey: 9 scrolled first, 1 album (10/15). In MVP testing, 3 of 5 previously found old photos via timeline scrolling and 2 via albums (0 via search!). Browse is the real workaround.

Recovery Scoped Out from Formula

Recovery is only 0.7% of engine failures and 1 in survey. 77.1% break before recovery is reached. It ships only as a lightweight safety net and is tracked as a diagnostic.

Strategic Metric Takeaways

- 1. Where Google Invested: Ask Photos, the hybrid router, and the Classic/AI toggle focused on prompt interpretation and latency, but left episode grouping and event-relative time unsolved.
- 2. The 77.1% Concentration: Interpretation (35.4%) and Surfacing (41.7%) represent 111 of 144 failures. Solving these two unlocks the vast majority of lost retrievals.
- 3. The Expression Disagreement: Engine reports 1.4% vs Survey 20% (3/15). However, 2 of the 3 survey non-typers were timeline scrollers. The moment view serves them directly.

The engine turns 85,140 public posts into 144 retrieval attempts, each recording what was remembered, what was forgotten, and where search broke



Extracted Field in Schema	The Brief's Underlying Question Answered	Reliability Check
asset_type	What kinds of old photos do users struggle to retrieve?	Validated in Gate B
cues_retained (14 cue types)	What do people actually remember about the photo?	Jaccard 0.557
cues_lost (date, place, album, words)	What have they forgotten when search fails?	Jaccard 0.673
query_verbatim · search_mode	How do they search when memory is incomplete?	Quote verify 85.2%
failure_stage (5 stages)	Where does retrieval break? (Part 3 decomposition)	κ 0.509 (Moderate)

Why This Is More Than Summarisation

1. Structure, not sentiment: Every review was converted into a structured, typed record with fixed vocabularies, not qualitative impression tags.

2. Pre-registered rules: Hypotheses H1–H5 and decision thresholds were frozen before data collection, preventing post-hoc confirmation bias.

3. The audit overruled the ranking: When the blind audit showed hypothesis agreement was low (Jaccard 0.347) while failure stage agreement was solid (κ 0.509), the engine demoted hypothesis ranking and pivoted to failure stages.

retrieval-discovery-engine.vercel.app ↗

People remember roughly when, but search needs the exact date, and the exact date is what they forget

The Brief's Four Questions Answered (Engine vs. Survey)

Question	Engine (144 Specific Attempts)	Survey (n=15 / 14 Dedup)
Photo Kinds	Personal photos 80 · multi-photo 26 · videos 16 · screenshots 7 · docs 3	All 3 cases of severe trouble were documents or medicine bills
Remembered	Roughly when 40 · object 17 · exact date 14 · text 12 · people 10	Roughly when 8 · object 7 · people with me 6
Forgotten	The date 37 · album 29 · exact words 10 · place 10	When it was taken 9 · words to search 9
Search Method	20 of 32 queries are 1 word ("dog"); 10 name a time ("Halloween 2024")	First move: scroll 9/15; own-app: 4 of 5 strip time to bare nouns

Opportunity Areas: Why O1 Leads (§B2)

O1 (Rough time or an event): 42 attempts, 20 not surfaced. *Core focus.*

O8 (No clue at all): 47 attempts. Larger, but users kept zero search clues. O1 is the largest group where user memory exists and search fails to use it.

Excluded: O7 (Exact date known, 14 — regular search works); O9 (Path moved by UI update, 12 — app design).

Pre-Registered Hypothesis Verdicts (Rule B)

H1 Episodic Time: SUPPORTED (Root Cause). Roughly when is #1 kept cue (40); exact date is #1 lost cue (37). The search engine demands what memory loses.

H2 Visual Recognition: SUPPORTED (Secondary). 41.7% never surfaced; 7 of 15 survey ended unsure; grouping into moments resolves recognition ambiguity.

H3 Dead-End Recovery: REFINED, NOT LEAD. Overruled by audit: agreement was weak (Jaccard 0.347), and reliable failure stage κ 0.509 shows cannot_refine is only 0.7%.

H4 Hinglish / Code-Mixed: UNTESTED IN ENGINE. Weak survey signal (2/15 survey; 1/6 testing). Retained as a deliberate design choice, not empirical finding.

H5 Content Not Indexed: WEAK VOLUME. Low volume (3 docs), but causes disproportionate real-world user harm.

H6 Path Changed: PRESENT, MINOR (8.3%). Post-hoc finding from navigation updates.

Most people scroll before they search, and nearly half end close but unsure

Research Methodology Rationale

Structured Questionnaire (n=15, 23–30 Sep): Replaced open interviews to eliminate unscorable anecdotes. Every response maps 1:1 onto the discovery engine vocabulary.

Why This Method: (1) 100% scoreable narratives (vs 8.6% in public posts); (2) Shared vocabulary with engine; (3) Rapid reach (15 participants vs 2 interview contacts).

Costs Acknowledged: Self-report rather than observation; convenience sample from network. 8 of 15 fit target segment.

Survey vs. Engine (Side-by-Side)

Metric	Engine (144)	Survey (15 / 14)
Most remembered	Roughly when (40)	Roughly when (8)
Most forgotten	The date (37)	When taken (9)
Top failure	Never surfaced (41.7%)	Never surfaced (5 / 4)
Ended unsure	Not measurable	7 of 15 (47%)
First move: scroll	15 workarounds	9 of 15 (60%)

Observed Real Retrieval Incidents (Survey Participants)

ID	Target Photo	What They Remembered	What They Typed	Break	Outcome
R04	Wedding photo	Roughly when, event, people	"wedding, pichle saal diwali"	Misread	Unsure
R02	Personal photo	Roughly when, event, object	"gym"	Eval fail	Unsure
R03	Personal photo	Roughly when, event	(Didn't know what to type; scrolled)	Expression	Found
R11	Personal photo	Roughly when, event	(Never typed; opened album)	Expression	Unsure
R06	Medicine bill	Object, text in bill	"medicine, bill"	Surfacing	Lost (re-got)

"wedding, pichle saal diwali" — R04 (Hinglish: a festival and relative year, not a calendar date)

"medicine, bill" — R06 (Never found; had to request document from clinic again)

Ask Photos Reality: 7 of 15 had never heard of it. All 4 who described a result said it showed "related photos, but not the one I wanted" and that they "could not tell why".

The segment is people who can place the moment only roughly, and search can't turn "roughly" into a date window

Target Segment Profile
Long-time Google Photos users searching for a personal moment they can place only roughly.

Defined by **cognitive memory state** (what they remember), not demographics: "around Diwali", "my sister's graduation", "last winter".

Segment Sizing Filter	Engine (144)	Survey (15)
Kept rough time or event (O1)	42 (29%)	8 (53%)
Target Segment (After Exclusions)	37 (26%)	8 (53%)
Personal photo or video	30 of 37	All 8
Large library (5,000+ items)	Not measured	5 of 8

Deliberate Exclusions: 14 who remembered exact date (standard search works) · 12 whose path moved in an update (H6 navigation) · Cloud sync / corruption issues.

The Root Cause: The Broken Link



"I'll type in something super simple, like 'Halloween 2024' and it seriously can't find anything?" — Play Store review

- How Retrieval Breaks for the Segment (37 Attempts):
- **Misinterpretation (12):** Query clue is misread; search fails to map the event to a date window.
 - **Surfacing Failure (18):** Returns an unranked grid across years; target photo never surfaces.
 - **Real-App Probe Verification (§G2):** 4 of 5 users compressed memory into bare nouns ("vacation", "gym"), and 0 of 5 found the photo!

People already scroll to "roughly when" by hand and still end unsure, so the product should take that step for them

LOCKED PROBLEM STATEMENT

"People remember when-ish and what happened; Google Photos indexes items and calendar dates. So the photo is in the library, the person can describe the moment but not the photo, and it never comes back."

The Workaround is the Design Blueprint

Engine: 15 of 21 workarounds are manual scrolling (7 of 10 failed): *"Every time I scrolled to the right time frame, it kept loading more pictures and moving up a couple of years ahead."*

Survey: Scrolling was the first move for 9 of 15. 5 of those 9 still ended unsure.

MVP Testing (§G2): When asked what actually found old photos previously: 3 timeline scroll, 2 album, **0 search!**

Insight: Users manually execute episodic retrieval by scrolling to a date region. The product should automate the date window and cluster the moments.

Solution Rationale & Why Not Ask Photos

Intelligence where it's needed: (1) Resolving clues into date spans; (2) Grouping photos into recognizable visual episodes. Runs entirely on existing device metadata.

Why Not Ask Photos: Ask Photos improved natural language routing and speed, but lacks editable clues, episode groupings, and explainable match ledgers. **Across survey (4/4) and MVP tests (3/3), all 7 users who tried Ask Photos got "related photos, but not the one" (0 found).**

Scoped Out with Evidence: Clarifying question (engine 1.4%; survey non-typers browse) · Full recovery agent (0.7% engine; survey 1/15) — replaced by 1-tap clue alternatives.

HOW THE THINKING EVOLVED (BRIEF PART 4 REQUIRED STEPS)

1. **Business Metric:** Per-search success → **Per-user URR** (the brief counts users)
2. **Product Outcomes:** Recovery agent → **5 stages + browse path** beside search
3. **AI Discovery:** Audit initially ranked H3 top → **Overtuned:** 77% fail before recovery
4. **Observed Behavior:** Users search and retry → **They scroll to a time by hand and end unsure**
5. **Problem Definition:** "Search is bad at old photos" → **Items vs. episodes mismatch**

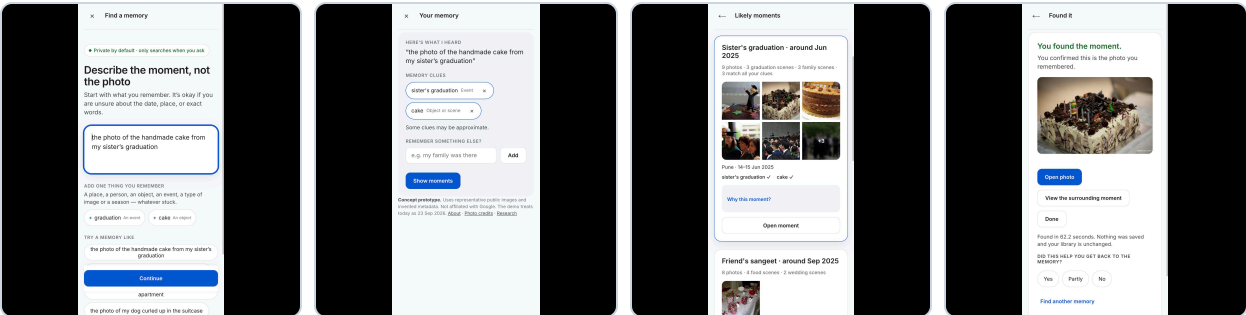
Memory Trails turns "roughly when and where" into moments you can recognise

Placement & Interaction Flow

Where it lives: Native feature inside Google Photos search. When plain search fails, *"Can't describe it?"* launches Memory Trails with query carried over.

Flow: Describe → Clue chips (editable, resolved date spans) → Up to 5 ranked moments → Moment inspection (before/after photos) → Confirm (*"That's the one"*).

Soft-Scoring Retrieval: Replaced hard date gating with exponential decay ($\exp(-\text{days_outside} / 7.0)$), ensuring photos near date boundaries are demoted, never hidden permanently.



memory-trails-v2.vercel.app

Shipped: fp16 ONNX, 1,282 photos

Empirical Evaluation Ladder (120 Tasks, 1,282 Photos)

Cue Dropout Level	What the Query Keeps	Plain CLIP	Memory Trails	Moment@5
L3 (Fully described)	Vague time + exact place + library word	0.172	0.962	0.933
L2 (Two vague cues)	Paraphrased content + 2 vague cues	0.209	0.276	0.333 ★
L1 (One vague cue)	Paraphrased content + 1 vague time cue	0.242	0.309	0.333
L0 (Content only)	Pure scene description, no metadata	0.312	0.312	0.267
Weighted by real memory	Weighted by real cue counts (§E1b)	0.259	0.334 (+0.075)	0.336 (+0.111)

Mandatory Disclosure: L3's 0.962 requires three cues, which only 4% (6 of 144) of real attempts kept. Weighted by real memory, the recall gain is **+0.075**, and the moment@5 gain **(+0.111)** proves the episode-grouping thesis over flat ranking.

"This MVP validates the memory-reentry interaction and recovery model using representative media. It does not validate production-scale Google Photos retrieval accuracy."

Testers found the moment in 5 of 6 cases and rated it easier than search, but compound dates need explicit control

ID	Own Google Photos Query → Result	Task 1 (Cat, Diwali 2025)	Task 2 (Dog, Diwali 2024)	Sureness	vs. GP	Intent
R01	"Wedding before COVID" → other years	Found, sure	Not found (year stuck)	5 / 5	5 / 5	Every time
R02	"gym" → couldn't tell which	Found, sure	Found, sure (1-tap alt)	4 / 5	4 / 5	When fails
R03	(Apple Photos user)	Found, sure	Found, sure (nearby months)	5 / 5	—	Every time
R04	"vacation" → other years	Found, sure	Found, sure (1-tap alt)	4 / 5	4 / 5	When fails
R05	"Wedding event" → couldn't tell which	Found, sure	Found, sure (clue edit)	4 / 5	4 / 5	Every time
R06	"Kerala photos" → couldn't tell which	Right Diwali, unsure photo	Found, sure (1-tap alt)	4 / 5	3 / 5	Every time

6 of 6
Segment fit (3 event, 3 roughly)

4.33 / 5
Mean Task 1 confidence

4.0 / 5
Ease vs GP (4 easier, 1 same)

0 of 5
Found photo on own Google Photos

User Quotes & Differentiators

- "Shows the moment, not a grid" — R02
- "Moving to nearby months, like scrolling but faster" — R03
- "Works without knowing what to type" — R04
- "Finds exact month & year... understands mix of Hindi and English" — R06

What Broke → Next Iteration (Brief Req.)

1. Compound relative time failed: R01 got stuck on "year before last" ("Understand 'pichle ke pichle saal'"). → Fix: Support compound year offsets and manual year dropdown edit.
2. Clue chip saliency: R04 didn't notice clue labels at first. → Fix: High-contrast badge: "Showing Diwali 2025 · Tap to change".
3. Retrieval speed: R05, R06 asked for faster loading. → Fix: Pre-cache moments and skeleton progressive image load.

Success is more users reaching the photo, and the biggest risk is that the win sits where memory is richest

Metrics Architecture: North Star URR

$$URR = Expression \times [1 - (1 - Interpretation \times Surfacing \times Recognition)^{\bar{n}}]$$

North Star: +8 percentage points lift in 28-day URR in user-level randomized experiment (6-month evaluation).

Why no Recovery term: Recovery is 0.7% of failures. It ships as a light safety net and is tracked as a diagnostic side-metric.

Leading & Guardrail Metrics (Live Events)

Metric	Instrumented Telemetry Event	Target / Guardrail
Entry rate	memory_reentry_started	≥12% of failed searches
Clue correction rate	chip_alternative_taken	15–25% (healthy agency)
Confirm-after-open	retrieval_confirmed / moment_opened	≥55% conversion
False confirmation ★	Confirmed photo ≠ Target photo	Guardrail: <3% (Stop launch)
P95 Latency	secondsToFirstMoment	Guardrail: <1,500ms

Risk Matrix & Mitigations (Part 8)

Identified Risk	Built Mitigation & Evidence
R1. Win concentrated where memory is richest (+0.79 at L3 vs +0.07 at L1)	Lean on moment recognition (moment@5 0.333 at L2 beats recall@20 0.276); measure L1 in live A/B.
R2. Own date filters hide photos (rigid boundaries cause not_surfaced)	Soft scoring built: exponential decay outside window lifted L3 recall 0.851 → 0.962.
R3. Index cannot read text in images (0.000 recall on receipts/notes)	Incorporate OCR embeddings and caption extraction in production roadmap.
R4. False confirmation (user selects wrong photo; worst failure)	Never auto-confirm; display before/after photos; testing showed 4.33/5 confidence.
R5. Ask Photos may already suffice (incumbent AI assistant)	7 of 7 users across survey and tests reported Ask Photos gave "related photos, but not the one".

Honest Limitations: Play Store 86.2% volume skew · Hinglish is a weak signal (2 in survey, 1 in testing) · URR baselines modelled · Ranking weights are heuristic · Survey (n=15) and testing (n=6) are unmoderated convenience samples.